

DSP Lab-1

In Lab Task

1. Effect of Sampling Rate

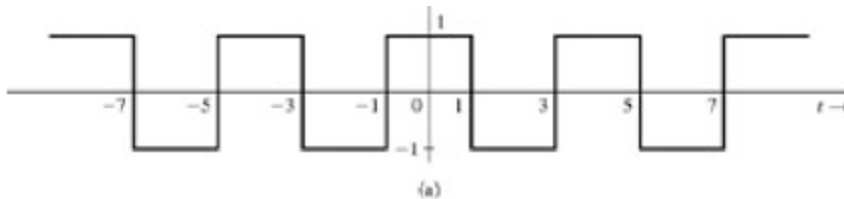
- Generate a sinusoidal signal $x(t) = \sin(2\pi f_0 t)$ where $f_0 = 500\text{Hz}$.
- Sample it at different rates: $f_s = 800\text{ Hz}, 1000\text{ Hz}, 2000\text{ Hz}$
- Reconstruct the signal using stem and plot in MATLAB.
- **Question:** What happens when $f_s < 2f_0$? Verify with spectrum plots.

2. Aliasing Demonstration

- Generate a continuous time 1 kHz sine wave call it $x_1(t)$. Sample at $f_s = 2\text{ kHz}$
- Generate a continuous time 3kHz sine wave call it $x_2(t)$. Sample at $f_s = 2\text{ kHz}$.
- Show by plotting that the discrete time signals are same even though continuous time signals are different.

3. Fourier Series for Continuous time signals

Consider the following periodic signal:



- Plot the original square wave in matlab
- Approximate it through Fourier Series using the first $N=5, 15, 50$ terms (or harmonics) by plotting for each case.
- Compare the approximation with the actual square wave and observe how the approximation improves when we take summation of more and more harmonics.

Post Lab Task

Undersampling Audio

- Take a short .wav file (voice) or record your own voice

- Resample the audio at different sampling rates (e.g., 44.1 kHz, 16 kHz, 8 kHz, 4 kHz, 2 kHz).
- Listen to each version and observe the quality.
- Plot the waveform and spectrum for at least two sampling rates (one high and one very low).
- Attach images of your plots with your answers.
- **Question:** Identify the lowest sampling rate at which the signal is still intelligible.